

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks: 50

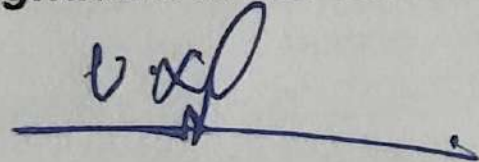
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I V. Yunnshankar ⁽¹⁹²⁵¹¹²⁶⁵⁾ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



1.) Event Communication Security :

a) Reconstruct the Encrypted message using shift 4 :

The appropriate classical encryption technique is the Caesar cipher.

In a Caesar cipher, every letter in the plaintext is shifted by a fixed number of positions in alphabet.

Here :

Shift = 4

The alphabet is :

plain : A B C D E F G H I J K L M N O P Q
 T U V W X Y Z

Shift +4 : E F G H I J K L M N O P Q R
 U V W X Y Z.

So :

① A → E

② B → F

③ C → G

④ D → H

⑤ E → I

⑥ F → J

⑦ M → Q

⑧ N → R

⑨ O → S

⑩ T → X

for 1 (a)

... = Q11X EX RSSR.

The

Encrypting Each character :

MEET AT NOON.

Now apply +4 to every alphabetic character

position	plaintext	+4 shift	ciphertext.
1	M	M → N → O → P	Q
2	E	E → F → G → H → I	I
3	T	T → U → V → W → X	X
4	A	A → B → C → D → E	E
5	T	T → U → V → W → X	X
6	N	N → O → P → Q → R	R
7	O	O → P → Q → R → S	S
8	O	O → P → Q → R → S	S
9	N	N → O → P → Q → R	R
10	N	N → O → P → Q → R	R

Spaces are not shifted.

Therefore :

plaintext : MEET AT NOON.

shift : +4

ciphertext : Q11X EX RSSR.

Final answer for 1 (a)

Encrypted message = G I I X E X R S S R

b) How does the authorised member retrieve the original message?

The authorised member must know the encryption rule which is:
Every rule letter was shifted 4 position forward.

Instead of shifting +4, the receiver shifts every letter -4

Decryption table:

Ciphertext	-4	Plaintext
G	M	M
I	E	E
I	E	E
X	T	T
E	A	A
X	T	T
R	N	N
S	O	O
S	O	O
R	N	N

Analysis : cipher is simple and easy
includes very limited.

Therefore

Ciphertext : QIIX EX RSSR

Shift : -4

Plaintext : MEET AT NOON

Mathematical representation :

Let each alphabet letter can be
represented by a number.

$$A = 0$$

$$B = 1$$

$$C = 2$$

$$\vdots$$

$$Z = 25$$

For encryption :

$$C = (P + 4) \bmod 26$$

where,

① P = Plaintext letter value.

② C = ciphertext letter value.

③ 4 = Shift value.

For decryption :

$$P = (C - 4) \bmod 26$$

The mod 26 operation is important.
The alphabet contains 26 letters. If shift
goes beyond Z, it wraps around.

Security Analysis :

The caesar cipher is simple and easy to implement but it provides very limited security.

There are only 25 possible non-zero shifts. An attacker who intercepts the ciphertext can try every possible shift until meaningful text appears.

For Example,

Q11X EX RSSR

An attacker could try :

It try,

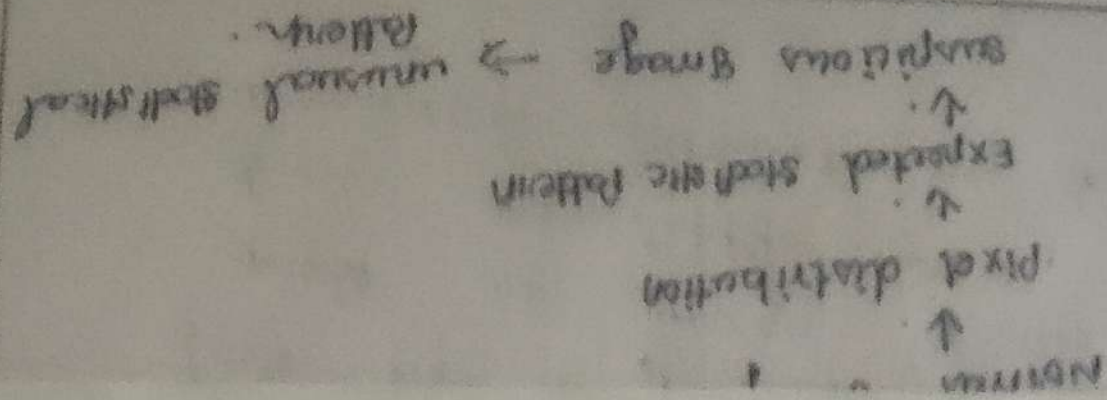
Shift - 1

Shift - 2

Shift - 3

Shift - 4 $\rightarrow$ MEET AT NOON.

So the cipher is suitable for understanding concept of classical encryption but it is not suitable for protecting sensitive modern communications.



Suspicious Digital Asset Investigation :

a) How can investigators detect hidden data?

i) visual inspection :

The first step is to examine the image carefully.

Investigators can look for,

- * unusual pixels.
- * unexpected color variations
- * distorted regions
- * strange patterns

For example two version appear identical
 Normal Logo.

↓
 compare

↓
 suspected logo

↓
 look for unusual pixel/color differences

2) compare image properties :

Investigate can examine properties

- * File size
- * Image dimensions
- * color depth

suppose a normal logo is :
 logo A → 150KB.

and the suspicious logo is :

logo B $\rightarrow$ 480KB.

The size difference does not prove the hidden data exist but it can be reason for further forensic examination.

3. Metadata analysis :

Digital images can be contain metadata. Investigate can examine metadata for :

- ① comments
- ② author information
- ③ software information
- ④ unusual fields

4. Statistical analysis :

This is a more technical detection method. Steganographic techniques may modify image pixels. Investigators can analyze statistical properties.

For example,

Normal Image

↓

Pixel distribution

↓

Expected statistical pattern

↓

suspicious image $\rightarrow$ unusual pattern

2) Robust Steganography :

A strong steganography technique
will be used to hide the information.

The distribution of pixel value is significantly different from what is expected. Investigators can investigate whether information has been embedded.

Use steganalysis tools :

Steganalysis is the process of analyzing digital media to determine whether hidden information may be present.

A specialized tools are,

- 1.) Inspect the file structure.
- 2.) Examine metadata
- 3.) analyze pixel patterns.
- 4.) extract data when possible.

Two Techniques to Improve Security :

- 1.) Encryption before hiding :

The confidential message should first be encrypted and then embed inside the image. If an attacker extracts the hidden data, it will be unreadable without the encryption.

2) Robust Steganography :

A strong steganography technique should be used to hide the information with minimum changes to the image. This makes the hidden data more difficult to detect or extract.

A strong Architecture is,

secret message

↓

Encryption

↓

cipher text

↓

Steganography

↓

Image file.

↓

Transmission

↓

Receiver

↓

Extract data

↓

Decrypt data → original

This provide defense in depth.

Conclusion $\frac{0}{0}$.

using encryption along with
steganography provides better security
Encryption protects the actual info
while steganography hides the existence
of communication.